

Manga109Script Dataset for Script-to-Layout Generation

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Abstract

001 A well-designed layout plays a crucial role in visual me-
002 dia, serving to both capture the audience’s attention and
003 facilitate information transfer. While this principle applies
004 across various media, manga has a unique layout that re-
005 flects its narrative structure and is now a widely applied
006 medium in fields beyond entertainment, such as education
007 and advertising. In manga production, layouts are crafted
008 based on a narrative script; consequently, the layout is a
009 key component of visual direction, where the arrangement
010 of panels and characters is constructed to effectively convey
011 the narrative and enhance dramatic effect. Although layout
012 generation has seen progress in domains such as Web UIs
013 and posters that prioritize visual clarity, there has been lit-
014 tle exploration into script-based layout generation, largely
015 due to the lack of data. In this study, we created script
016 data for approximately 20,000 pages from the Manga109
017 dataset and built the Manga109Script dataset linking scripts
018 to manga layouts. Leveraging this dataset, we developed
019 manga layout generation models that directly take scripts
020 as input, employing various LLMs as base models. Our
021 comprehensive evaluation shows that incorporating script
022 information demonstrably improves layout generation qual-
023 ity compared to existing methods. These results establish an
024 effective initial baseline for future research in script-based
025 layout generation, which we term Script2Layout. As such,
026 Script2Layout stands as a key enabling technology for appli-
027 cations designed to translate narratives into visual media,
028 as exemplified by manga.

029 1. Introduction

030 Storytelling takes many forms, from text-focused novels to
031 visually-driven animation. Among these, manga has devel-
032 oped a unique expressive form that divides a flat surface
033 into panels containing characters and dialogue. In typical
034 manga production, the process of translating a narrative into
035 a visual layout occurs in a preliminary draft called a *name*.
036 The layout therefore embodies critical directorial decisions.

Its composition, particularly the positioning of panels, char- 037
acters, and speech balloons, is a deliberate choice designed 038
to structure the visual narrative and ensure the story is com- 039
municated clearly and with intended emotional weight [9]. 040

In contrast to this narrative-centric process, the field of 041
layout generation has seen development, which has primarily 042
focused on functional applications like Web UIs [20] and 043
posters [10, 43], aiming to generate visually compelling lay- 044
outs that ensure clear and efficient information transfer [39]. 045
However, layouts that combine storytelling and dramatic 046
effects, like manga, are under-researched. The absence of 047
data linking stories to layouts is a key reason for this gap. 048
Script data are often produced internally and rarely released 049
publicly, making it difficult to obtain. This data scarcity has 050
been the primary barrier to developing assistive applications 051
for this narrative-driven creative domain. 052

In this study, we utilized Vision Large Language Models 053
(VLLMs) to create Manga109Script, a script dataset for ap- 054
proximately 20,000 manga pages based on the Manga109 055
dataset [1, 28]. Recent advances in VLLMs have enabled 056
high-precision extraction of textual information from im- 057
ages [51]. This capability has significantly contributed to 058
research in Visual Storytelling, which extracts narratives 059
from image sequences [26, 30]. Despite advancements in 060
VLLMs, understanding content across multiple panels re- 061
mains challenging [15, 44]. To mitigate VLLMs’ limitations, 062
we applied Visual Prompting [40] with the Manga109 dataset 063
annotations and employed a rule-based algorithm designed 064
for panel reading order [35], to generate high-quality narra- 065
tive script data. 066

Our scripts follow standard script formatting conventions, 067
including *scene headings* indicating location, *action descrip-* 068
tions detailing character movements and scenery, and *dia-* 069
logue (Fig. 1). We consulted with manga experts to con- 070
firm the format fits professional workflows. The generated 071
script’s quality was also evaluated by humans. 072

To demonstrate the practical utility of our dataset for such 073
creative applications, we developed manga layout generation 074
models that take scripts as direct input. We comprehensively 075
evaluated these models against existing layout generation 076

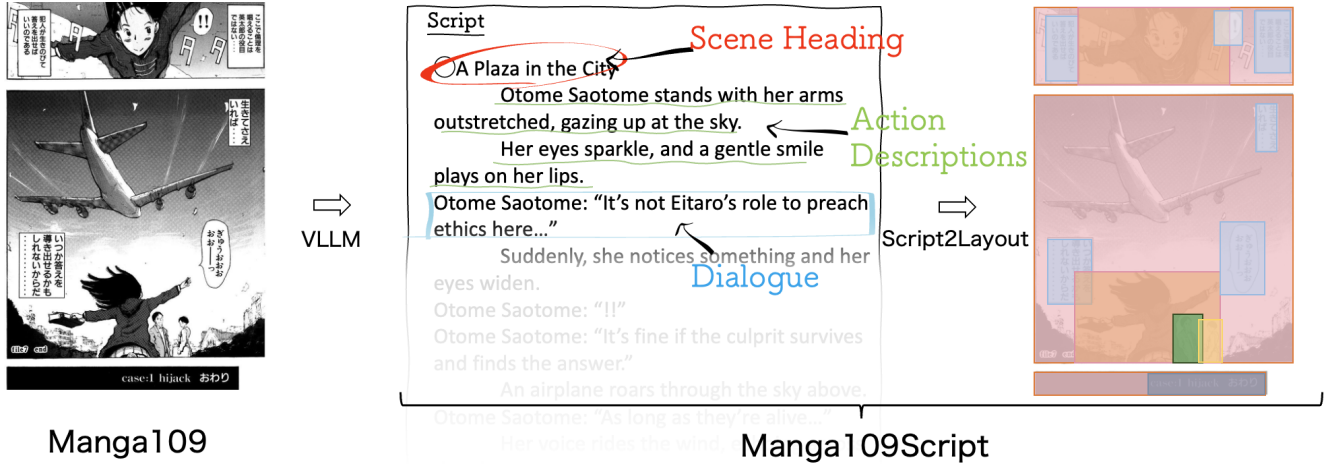


Figure 1. Overview of the Script2Layout task. From left to right: (1) a manga page from the Manga109 dataset; (2) its corresponding script, generated using a vision large language model (VLLM) with visual prompting and speaker-aware dialogue sequencing; and (3) the layout predicted by a Script2Layout model based on the generated script. The script follows standard manga production conventions and includes scene headings, action descriptions, and dialogue. This setup allows layout models to make use of narrative structure for generating more coherent and expressive layouts. (Source: *Hanzai Kousyounin Minegishi Eitarou* by Takashi Kii)

methods and assessed the impact of script input to establish a baseline for the Script2Layout task. Dataset, and code will be released upon publication.

Our contribution is threefold:

1. We introduce an annotation-assisted method to generate script data from manga images using VLLMs, validated by human evaluators.
2. We publish the first professional script-format manga storytelling dataset, covering 20,000 pages.
3. We establish robust baselines for the novel Script2Layout task, demonstrating for the first time how script influences manga layout.

2. Related Work

2.1. Layout Generation

Layouts crucially impact viewers' comprehension and perception of visual content. The pursuit of excellent layouts across various media has driven research in layout generation models [13, 38, 39]. Previous research primarily targets layouts emphasizing element visibility, such as mobile UIs [7, 17], posters [10, 37, 43], and PDFs [17, 52].

To meet the demands of practical applications and diverse design workflows, various constraint-based generation approaches have been developed. These approaches include generation conditioned solely on element types [18, 23], on both element types and size [18, 22], and relational constraints between elements [21, 23]. There are also interactive variations, such as layout completion that infers missing elements from partial layouts [11], and layout refinement that improves existing layouts [34]. Layout generation with fixed background images as a condition is called content-aware layout generation, which considers harmony with the back-

ground [12, 37].

Model development under these various constraints has enabled more accurate layout generation. Layout generation models have evolved from early GANs [24] and VAEs [2, 19] to Transformers [18] and diffusion models [4, 16, 50]. Recent studies have also explored the use of LLMs for this task [27, 43].

Despite these technical advancements, generating layout from narrative context is still unexplored. This challenge arises from dataset scarcity and the difficulty of converting narrative dynamics into visual layouts. Modern LLMs are now capable of interpreting complex contexts and nuances effectively [32]. Leveraging these capabilities, we propose Script2Layout, a task to generate layouts directly from scripts. We evaluate our approach against state-of-the-art methods such as LayoutDM [16], LayoutFormer++ [18], and LayoutPrompter [27], thereby demonstrating both its effectiveness and innovative potential.

2.2. Manga Understanding as Visual Storytelling

High-quality script data is essential for the Script2Layout task, and its construction requires advanced techniques for narrative extraction from images.

Visual storytelling, the task of generating narratives from image sequences, was pioneered by Huang [14]. The field evolved from early models using Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) [49] to later methods using hierarchical attention [29], scene graphs [46], and causal reasoning [41]. These techniques move beyond simple image recognition to capture inter-image relationships and ensure narrative coherence.

Recent advances in VLLMs like GPT-4o [47] have enhanced detailed context extraction from images, creating

new opportunities in Visual Storytelling [48]. However, applying VLLMs to manga reduces performance due to a domain gap from its distinctive visual language, *e.g.*, black-and-white art, unique reading orders, and discrete time and space within panels. These distinctive features make manga understanding a complex task composed of multiple intertwined subtasks. As illustrated in the CoMix benchmark [45], these include panel segmentation, dialogue ordering, and character re-identification across panels [35]. Character re-identification is particularly challenging, with GPT-4 achieving less than 50% accuracy [45].

To circumvent these challenges, we leverage solutions to key subtasks, providing the model with pre-processed inputs like ordered dialogue with speaker attribution [25] and annotated character regions [40]. This design allows the model to bypass low-level visual parsing and focus on generating high-quality narrative scripts, leveraging its core vision-language capabilities.

While datasets like Manga109Story [5] provide prose narratives, our primary aim is a *script-format* dataset. Accordingly, we developed Manga109Script with professional creators to fit conventional manga production workflows.

3. Manga109Script

Manga109Script is a script dataset corresponding to 19,555 manga pages from the Manga109 dataset [1, 28]. Manga109 consists of 109 commercial Japanese manga works published between the 1970s and 2010s, ensuring substantial variety in style and content. Manga109Script reformats the original two-page spreads into single pages and links each page to its corresponding script data.

The script format follows a style commonly used in Japanese manga production. Each script consists of three main components: (1) Scene Headings, marked with a leading “○” to denote location, (2) Action Descriptions, written with a three-character indent to describe characters’ actions and environmental details, and (3) Dialogue, where the speaker’s name is followed by their speech enclosed in quotation marks. These components are illustrated in the center of Figure 1.

We adopted this script format because manga has primarily developed within Japan, and all manga titles in this study were originally created through Japanese production processes. This ensures the dataset aligns with actual manga creation practices.

3.1. Data Construction Process

To construct Manga109Script, we employed GPT-4o via API to generate page-level scripts from manga images in the Manga109 dataset. To support the model’s understanding of character identity and dialogue context, we introduced two key inputs: (1) visual prompting, where the face of each character in the image was enclosed with a uniquely colored

circle [40], and (2) pre-extracted dialogue annotations, including the textual content of each speech balloon and its associated speaker’s name [25]. Dialogue order was pre-determined using the Magi algorithm [35], which provides a reliable rule-based ordering of panels and speech balloons. Without this step, GPT-4.1 alone achieved exact ordering on only 26 out of 109 sampled pages, whereas the Magi-based ordering reached 96 out of 109 pages, demonstrating the necessity of algorithmic ordering prior to script generation.

To further address the challenge of recognizing continuous narratives across multiple pages, we structured a unique structured interaction process with GPT-4o consisting of the following three components for each target page i :

- $\alpha(i)$: **Initial comprehension prompting**

We provided the full dialogue from the episode that includes page i and simulated an affirmative response from the model to promote narrative understanding.

Input: entire dialogue of the episode

Task: comprehend the episode

Agent: comprehension complete

- $\beta(i)$: **Script generation for the i -th page**

We provided GPT-4o with the processed image of page i (with visual prompting), the dialogue on page i , and the summary generated in step $\gamma(i-1)$, covering the story up to page $i-1$. The model was then instructed to generate the script for page i in the Japanese screenwriting format.

Input: page i image and dialogue,
and prior summary from $\gamma(i-1)$

Task: generate script for page i

Agent: script generation complete

- $\gamma(i)$: **Summarization up to the i -th page**

We then prompted the model to summarize the story up to page i to generate the summary input for $\beta(i+1)$.

Task: based on interaction history,
summarize the story so far

Agent: summarization complete

To generate the script for page i , we executed the sequence:

$$\alpha(i) \rightarrow \beta(i-1) \rightarrow \beta(i).$$

Here, the arrow ($\rightarrow$) indicates the chronological order in which each prompt was issued to GPT-4o.

To generate the summary of the story up to page i , we extended the sequence as follows:

$$\alpha(i) \rightarrow \beta(i-1) \rightarrow \beta(i) \rightarrow \gamma(i).$$

Note that for $i = 1$, these sequences become $\alpha(1) \rightarrow \beta(1)$ and $\alpha(1) \rightarrow \beta(1) \rightarrow \gamma(1)$, where $\beta(1)$ does not include a summary input.

Table 1. Statistics of Manga109Script. Distinct character types are counted by character identity per page, and distinct speaker types are counted by speaker identity per page.

Element	Count	Avg./page
<i>Layout Elements</i>		
Pages	19,555	—
Panels	103,850	5.31
Character instances	157,865	8.70
Distinct character types	60,812	3.11
Dialogue balloons	146,549	7.49
Distinct speaker types	55,778	2.85
<i>Script Elements</i>		
Scene headings	26,931	1.38
Action descriptions	138,890	7.04
Dialogues	134,901	7.10
Japanese characters (text)	6,703,063	342.78

By iteratively applying this series of interactions from $i = 1$ to the final page of each manga title, we constructed the full Manga109Script dataset. The dataset statistics are summarized in Table 1.

3.2. Quality Verification

Unlike general image captioning datasets, our scripts lack ground truth. Moreover, the script format provides contextual coherence rather than word overlap. Thus, we used human evaluation with specific metrics to verify script quality.

Evaluation items included: (1) *Scene Heading Recall*, determining whether locations depicted in the manga were correctly identified in script scene headings; (2) *Scene Heading Precision*, determining whether locations mentioned in script scene headings actually appeared in the manga; (3) *Action Description Precision*, determining whether action descriptions generated for a target page actually match that page when mixed with randomly selected action descriptions from other pages; (4) *Out-of-Context Action Description Detection Accuracy*, determining whether action descriptions from other pages could be correctly identified as not belonging to the target page; and (5) *Dialogue Recall*, determining whether dialogue appearing in the manga was accurately transcribed in the script.

We randomly selected approximately 10 consecutive pages from each work to maintain story continuity, totaling 1,000 pages. The 1,000 pages were divided into two sets of 500 pages each, with six evaluators each reviewing one set, resulting in three evaluations per page. These evaluators received fair compensation.

We calculated each evaluation metric by averaging correct judgments across all elements: Scene Heading Recall (0.793), Scene Heading Precision (0.838), Action Description Precision (0.664), Out-of-Context Action Description

Detection Accuracy (0.910), and Dialogue Recall (0.988).

4. Experiments

The primary goal of our experiments is to evaluate the effectiveness of using scripts as input for manga layout generation and to establish a baseline for Script2Layout. Using Manga109Script, we compare Script2Layout models that are fine-tuned (FT) LLMs generating layout coordinates conditioned on script input, with existing methods [16, 18, 27] under multiple constraint settings, including the presence or absence of script input.

4.1. Problem Formulation

We formulate manga layout generation as the task of predicting spatial positions for a given list of element categories. A layout ℓ consisting of $N \in \mathbb{N}$ elements is represented as $\ell = \{(c_1, \mathbf{b}_1), \dots, (c_N, \mathbf{b}_N)\}$, where c_i denotes the category of the i -th element and $\mathbf{b}_i$ is its corresponding bounding box. In our experiments, we consider three types of element categories: *characters*, *speech balloons*, and *panels*. Each bounding box $\mathbf{b}_i$ is quantized on a 32×32 grid. Given the number of elements N and their categories $\{c_i\}_{i=1}^N$, the task is to predict their spatial positions $\{\mathbf{b}_i\}_{i=1}^N$ on the manga page.

This formulation reflects real-world manga production workflows, where artists compose layouts (known as “name” in Japanese production) based on a predefined script that specifies the categories of elements to be placed on each page, prior to any visual content creation. This formulation also bridges a critical gap between traditional layout generation models, which cannot directly accept script input, and Script2Layout models, which are designed to condition layout prediction on narrative scripts.

Under this formulation, we first describe how existing layout generation methods are applied to this task. We then introduce our Script2Layout models, highlighting the differences in how they utilize script information.

Existing Methods. The input and output formats of conventional methods are illustrated in the bottom part of Figure 2. Conventional layout generation methods require element categories to be predefined. In our experiments, categories are based on the three main types (*characters*, *speech balloons*, and *panels*) and are further distinguished on a per-page basis. Specifically, we assign category as $c_i \in \{\text{Chara}_i \mid i = 1, \dots, 25\} \cup \{\text{Speech}_i \mid i = 0, \dots, 25\} \cup \{\text{Panel}\}$. Here, Chara_i denotes the i -th character instance on a page, Speech_i represents a speech balloon spoken by Chara_i (or narration when $i = 0$). The number 25 was chosen to provide a buffer, as the maximum number of distinct characters in a single page of Manga109Script is 22. Based on this formulation, we evaluate three representative layout generation methods: LayoutDM [16], Layout-

Script2Layout

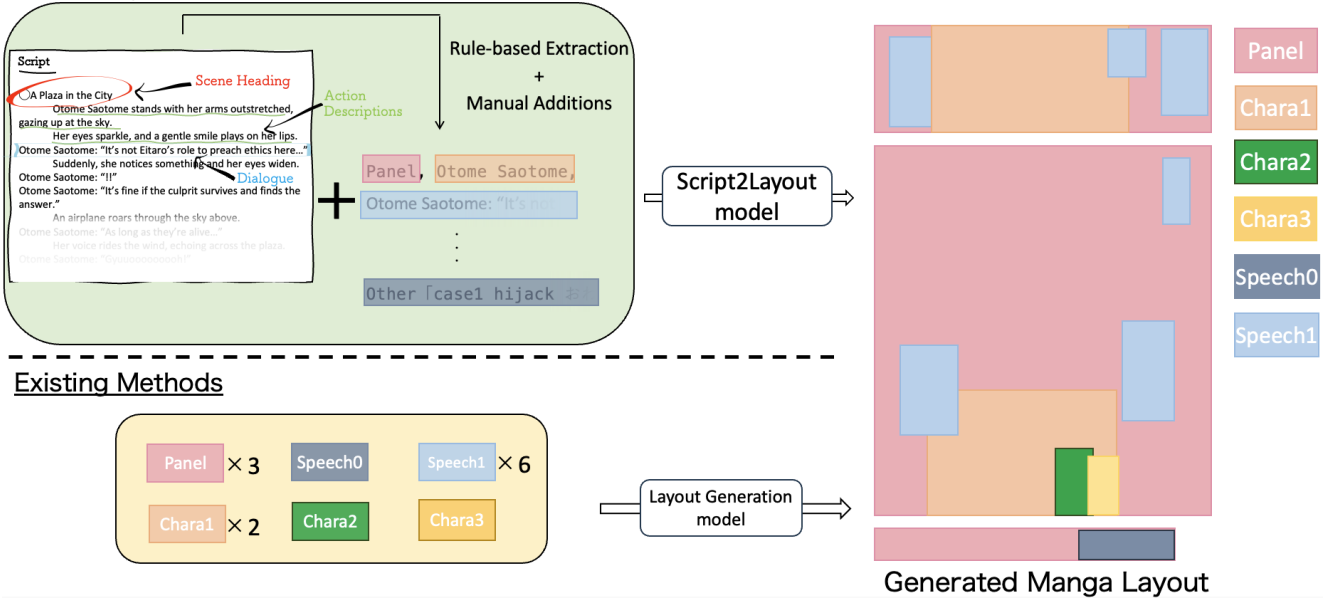


Figure 2. Comparison of manga layout generation approaches. Top: Our Script2Layout method takes natural language scripts and extracted layout elements (character names, dialogue, panels) as input to generate layouts via bounding box prediction. Bottom: Traditional methods require predefined category labels (e.g., Speech1, Chara2) as input. Unlike conventional approaches that rely on structured element types, Script2Layout enables direct layout generation from raw narrative scripts conditioned on storytelling intent.

Former++ [18], and LayoutPrompter [27]. The first two models are trained on the manga layout task. LayoutPrompter, in contrast, is a prompt-based system that does not involve model training; we implemented it using the GPT-4o API. Although the system is capable of accepting script inputs, we did not provide script information in our experiments due to input length limitations. Consequently, LayoutPrompter was evaluated under the same conditions as other non-script-based methods. We adopt generation conditioned on types (Gen-T) as the default setting throughout this study. In this setting, each element’s category is known. In addition, we evaluate the existing methods under two extended constraint settings:

- **Gen-TS (Generation conditioned on types and sizes):** Each element’s size (i.e., width and height) is given in addition to its category.
- **Gen-R (Generation conditioned on relationships):** Pairwise spatial relations (e.g., "element A is above element B") between layout elements are provided as constraints.

Script2Layout Models Under the above formulation, we use the following LLMs as base for Script2Layout models: T5 [33, 42], Swallow-13B [8], Sarashina-13 [36], and DeepSeek-R1-Distill-Qwen-14B-Japanese (DeepSeek-R1-J) [6], selected for their strong Japanese language capabilities and fine-tuning flexibility. The input and output formats of Script2Layout models are illustrated in the upper part of Figure 2. Unlike conventional methods that require predefined category labels, Script2Layout models can directly process

linguistic content. They take as input the raw content of layout elements, such as character names and speech texts, extracted from the script. These contents are passed to the tokenizer of each LLM without assigning predefined category labels, allowing the model to interpret them directly based on its pretrained knowledge. The model then outputs a quantized bounding box $\mathbf{b}_i = ((x_1, y_1), (x_2, y_2))$ for each element. In addition, we design input formats that allow the model to incorporate the script itself. Specifically, we consider the following three variants:

- **None:** No script is included in the input.
- **Plain:** Raw script text is included as input.
- **Panel-Inserted:** Script input is augmented with explicit panel segmentation markers, indicating how the script content is divided across individual panels.

In addition, we further evaluated Script2Layout models under the Gen-TS setting, combining it with the **None** and **Plain** input conditions. This additional experiment aims to assess the extent to which script information contributes to layout prediction when the sizes of all elements are provided in advance.

4.2. Dataset Split

For training and evaluation, we filtered out 90 pages from Manga109Script that contained more than 50 layout elements, as such cases can cause combinatorial explosion in constraint-based models (particularly under Gen-R). The remaining 19,465 pages were used in our experiments.

We split the data by manga title to avoid title-level leakage across splits. The split maintains an 8:1:1 ratio, with 15,333 pages for training, 2,007 for validation, and 2,125 for testing. The training set was used to fine-tune LLMs for Script2Layout and to train conventional baseline models from randomly initialized parameters. The validation set was used for early stopping, and the test set for final performance evaluation.

4.3. Evaluation Metrics

Given the diverse possibilities of manga layouts, which make evaluation difficult, we follow established metrics to assess their fidelity, spatial consistency, and readability: Fréchet Inception Distance (FID), mean Intersection over Union (mIoU), Overlap of Speech Balloons (Ovp.) and LTSim-MMD [31]

FID FID measures the similarity between distributions of real and generated layouts in a learned feature space. Following LayoutDM [16], we use FIDNetV3 as the feature extractor, which has been shown to be better suited for layout features than traditional conventional method. For Script2Layout, FID is computed between generated layouts and real layouts in the test set. As a reference, we also compute the FID between layouts in the validation and test sets and report it as “Real data.”

mIoU Mean Intersection over Union (mIoU) is computed by greedily matching each generated element with the ground-truth element of the same category that yields the highest IoU. This metric, commonly used in prior work [3], measures category-specific spatial accuracy and is reported as the average over all test samples.

Overlap of Speech Balloons We compute readability loss by first summing the pairwise IoUs of all balloons on each page, and then averaging this value across the dataset. In manga, overlaps between characters or between characters and speech balloons are often used as stylistic devices, but balloon-balloon overlaps are generally detrimental to readability. Therefore, we adapt the commonly used Overlap metric [24] to penalize only overlaps between speech balloons.

LTSim-MMD We also adopt LTSim-MMD [31], which is based on optimal transport and evaluates loose geometric consistency through many-to-many relationships. This allows us to measure collection-level similarity to the ground truth, and it is expected to capture the plausibility of manga layouts.

Qualitative Evaluation We qualitatively compare generated layouts with ground truth by visualizing outputs from different models, assessing how well they capture spatial structure and benefit from script information.

4.4. Implementation Details

Our Script2Layout models were fine-tuned using the Low-Rank Adaptation (LoRA) technique. We utilized the Hugging Face transformers and peft libraries for all implementations. All training and inference were conducted on a single NVIDIA H100 GPU. A comprehensive list of hyperparameters, including the LoRA configuration and training parameters, is provided in the Appendix.

4.5. Evaluations and Discussions

Quantitative results under Gen-T As shown in Table 2, scripts improve layouts for most models. From None to Plain, FID improves for 3/4 models and mIoU for 4/4, with mean relative changes of -10.3% (FID) and +9.0% (mIoU). LTSim-MMD also shows consistent gains, improving in all 4/4 models under this transition. From None to Panel-Inserted, FID improves in 2/4 models, mIoU in 4/4, Ovp. in 3/4, and LTSim-MMD in 3/4. These patterns suggest that scripts act as effective priors for manga layout generation. Character descriptions and speaker attribution likely serve as anchors for determining where speakers and speech balloons should appear, thereby helping to reduce Ovp., improve mIoU, and yield layouts that are semantically closer to ground truth, as captured by LTSim-MMD. In fact, under the Panel-Inserted condition, most models achieve improvements that surpass existing layout methods for these metrics. Moreover, we observe a strong dependence on model capacity. The impact of script input depends strongly on the underlying LLM’s capacity. Weaker models such as T5 sometimes degrade on certain metrics when scripts are provided, suggesting that they underutilize the richer input. In contrast, stronger models like DeepSeek consistently benefit, showing improvements in nearly all metrics; under the Panel-Inserted condition, DeepSeek surpasses existing layout models across all evaluation measures. These results indicate that rich script cues are most effective when the base LLM is sufficiently capable, whereas limited models may be hindered by the additional complexity. As LLM capabilities continue to advance, script conditioning is expected to become increasingly useful in practice for producing more coherent and readable manga layouts.

Qualitative Results Figure 3 shows that models conditioned on scripts produce layouts closer to the real pages. In the top row, the target page features a dramatic close-up of a character pointing a gun at his own temple. Script2Layout models with script input consistently place large character regions at the top-left, mirroring the original page structure. This alignment likely reflects textual cues such as “Norman stands frozen in shock. His eyes are wide open, with sweat beading on his forehead,” which may have guided the models to allocate spatial prominence to the main character. The bottom row shows a page with one dominant character with

Table 2. Results under the Gen-T setting. Script2Layout models, fine-tuned (FT) LLMs for layout prediction, are evaluated with three script input types (None, Plain, Panel). They generally outperform existing layout models, demonstrating the effectiveness of script input.

Script		None				Plain				Panel-Inserted			
Model	Metric	FID↓	mIoU↑	Ovp. ↓	LTSim -MMD ($\times 10^{-2}$)↓	FID↓	mIoU↑	Ovp. ↓	LTSim -MMD ($\times 10^{-2}$)↓	FID↓	mIoU↑	Ovp. ↓	LTSim -MMD ($\times 10^{-2}$)↓
Layout Generation Models													
LayoutDM		22.428	0.161	0.084	0.763	—	—	—	—	—	—	—	—
LayoutFormer++		28.402	0.140	0.203	2.236	—	—	—	—	—	—	—	—
LayoutPrompter		133.502	0.138	<u>0.020</u>	1.230	—	—	—	—	—	—	—	—
Script2Layout Models													
T5 (FT)		29.549	0.201	0.061	0.587	28.744	0.210	0.064	0.505	31.287	<u>0.204</u>	0.180	1.147
Swallow (FT)		35.845	0.156	0.021	0.921	18.415	0.182	0.026	0.429	<u>21.557</u>	0.184	0.013	0.394
Sarashina (FT)		20.377	<u>0.192</u>	0.028	<u>0.398</u>	24.191	<u>0.207</u>	<u>0.012</u>	<u>0.393</u>	25.127	0.213	<u>0.009</u>	<u>0.382</u>
DeepSeek-R1-J (FT)		24.618	0.174	0.007	0.383	<u>22.502</u>	0.186	0.005	0.359	21.094	0.184	0.004	0.291
Real data		4.780	—	0.002	—	4.780	—	0.002	—	4.780	—	0.002	—



Figure 3. Qualitative comparison of generated layouts. Each row shows one example: the real manga layout (left), followed by predictions from existing methods and Script2Layout models with different script input settings (None, Plain, Panel-Inserted). Script2Layout models generally produce layouts closer to the real ones, particularly when script information is provided. (Source: *Doll Gun* by Ryusei Deguchi, *Touta Mairimasu* by Shinji Saijo)

a bamboo sword. Script2Layout models with script input place a large character region centrally with a speech balloon at the upper right, mimicking the original composition. The script line "Ayumi Maniwa holds her bamboo sword at the ready, standing with a dignified posture" likely encouraged this semantically meaningful layout. Overall, these examples suggest that leveraging script information is a promising approach for generating layouts with greater narrative coherence and expressiveness.

Constraint Analysis Relational constraints (Gen-R), defining explicit pairwise element relationships, offered little to no performance gain over the Gen-T baseline for conventional models (Table 3). This suggests that simple pairwise relations are less effective than the rich, contextual guidance provided by a narrative script. In contrast, providing element sizes (Gen-TS) dramatically boosted performance across all models, both conventional and our own (Table 3 and 4). For instance, LayoutFormer++'s mIoU jumped

Table 3. Evaluation results of existing layout models under different constraint settings (Constr.).

Model	Constr.	FID ↓	mIoU ↑	Ovp. ↓	LTSim
					-MMD ($\times 10^{-2}$)↓
LayoutDM	Gen-T	22.428	0.161	0.084	0.763
	Gen-TS	20.514	0.271	0.101	0.165
	Gen-R	<u>20.864</u>	0.053	0.167	8.797
LayoutFormer++	Gen-T	28.402	0.140	<u>0.203</u>	2.236
	Gen-TS	17.487	0.208	0.079	0.244
	Gen-R	<u>26.848</u>	<u>0.158</u>	0.261	<u>2.187</u>
LayoutPrompter	Gen-T	133.503	0.138	0.020	1.230
	Gen-TS	58.993	0.148	0.037	0.855
	Gen-R	<u>130.502</u>	0.097	<u>0.024</u>	4.036

Table 4. Performance of Script2Layout models (FT LLMs) under Gen-T and Gen-TS with/without script input.

Model	Constr.	Script	FID ↓	mIoU ↑	Ovp. ↓	LTSim
						-MMD ($\times 10^{-2}$)↓
T5	Gen-T	None	29.549	0.201	0.061	0.587
		Plain	28.744	0.210	0.064	0.505
	Gen-TS	None	19.445	<u>0.379</u>	<u>0.053</u>	0.097
		Plain	<u>19.734</u>	0.394	0.041	<u>0.136</u>
Swallow	Gen-T	None	35.845	0.156	<u>0.021</u>	0.921
		Plain	18.415	0.182	0.026	0.429
	Gen-TS	None	20.650	0.337	0.020	0.046
		Plain	<u>19.946</u>	<u>0.334</u>	<u>0.021</u>	<u>0.057</u>
Sarashina	Gen-T	None	20.377	0.192	0.028	0.398
		Plain	24.191	0.207	0.012	0.393
	Gen-TS	None	18.165	0.358	<u>0.017</u>	<u>0.090</u>
		Plain	<u>18.783</u>	<u>0.352</u>	0.020	0.087
DeepSeek	Gen-T	None	24.618	0.174	0.007	0.383
		Plain	22.502	0.186	0.005	0.359
	Gen-TS	None	<u>20.769</u>	<u>0.381</u>	0.007	0.018
		Plain	20.013	0.383	<u>0.006</u>	<u>0.019</u>

from 0.140 to 0.208 and its LTSim-MMD dropped from 2.236×10^{-2} to 0.244×10^{-2} , while T5 model reached 0.394 in mIoU and improved its LTSim-MMD from 0.587×10^{-2} to 0.097×10^{-2} , affirming the critical role of size information. On the other hand, under the Gen-TS condition, where size information already provides strong improvements, the addition of script input did not consistently lead to further gains. This suggests that once geometric constraints sufficiently boost performance, it becomes more challenging for semantic cues to contribute additional improvements.

Failure Cases and Limitations Despite the promising results, our approach has several limitations. First, its reliance on bounding boxes prevents it from capturing the full geomet-

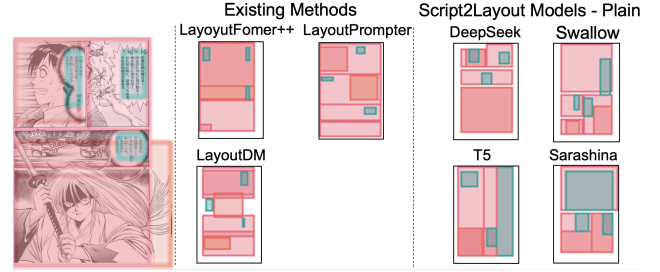


Figure 4. Failure cases of generated layouts. Each row shows examples: the real manga layout (left), followed by predictions from existing methods and Script2Layout models (Source: *Touta Mairimasu* by Shinji Saijo)

ric complexity of manga, such as slanted or non-rectangular panels. Second, because our models directly process the script’s textual content, they can generate disproportionately large speech balloons for long dialogues (Figure 4). Finally, our LLM-based approach incurs significant computational cost; processing scripts results in a per-sample generation time of approximately 7–12 seconds depending on script length slower than baselines like LayoutDM (approx. 0.66 s). Addressing these challenges, future work could also explore conditioning diffusion models on scripts to mitigate these limitations, particularly in efficiency.

5. Conclusion

This paper introduced Script2Layout, a framework for manga layout generation from scripts, and constructed Manga109Script, the first large-scale dataset for this task. Experiments showed that script conditioning improves performance over existing methods across multiple metrics. Limitations include that script cues did not always yield gains under strong constraint conditions and that our LLM-based approach incurs higher computational cost, resulting in slower generation compared to diffusion-based baselines. Future work includes combining size and script cues, exploring stronger base models, and assessing practical implementations. Beyond manga, these advances could contribute more broadly to story-to-visual content generation and narrative visualization applications.

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